BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE PROJECT

Executive Summary

The Mutual Fund Analytics Capstone Project was developed to analyze mutual fund industry trends, investor behavior, fund performance, and risk metrics using real-world financial datasets. The project implemented a complete data analytics pipeline consisting of data ingestion, cleaning, database design, exploratory data analysis (EDA), performance analytics, advanced risk analytics, and interactive dashboard development.

Python was used for data preprocessing, statistical analysis, and visualization. SQLite was used to build a structured analytical database. Power BI was used to create an interactive dashboard for business users. The final solution provides insights into mutual fund growth, SIP trends, fund performance, investor demographics, and portfolio risk characteristics.

1. Introduction

Background

The mutual fund industry has witnessed significant growth due to increasing investor participation, digital investment platforms, and awareness regarding systematic investment plans (SIPs). Asset Management Companies (AMCs) continuously launch new schemes to cater to different investor requirements.

Analyzing mutual fund data helps investors and fund managers understand:

- Industry growth trends
- Investor participation patterns

- Fund performance characteristics
- Risk-adjusted returns
- Portfolio diversification

Objectives

The primary objectives of this project were:

1. Build an end-to-end data analytics pipeline.
2. Design a relational database for mutual fund data.
3. Perform exploratory data analysis.
4. Evaluate fund performance using financial metrics.
5. Analyze investor behavior and SIP patterns.
6. Develop risk analytics models.
7. Build an interactive Power BI dashboard.
8. Generate actionable insights and recommendations.

2. Data Sources

The project used multiple datasets covering different aspects of the mutual fund ecosystem.

Datasets Used

Fund Master

Contains scheme information including:

- AMFI Code
- Fund Name
- Category
- Fund House

- Risk Level

NAV History

Contains historical Net Asset Value data for all schemes.

Investor Transactions

Contains:

- Investor ID
- Transaction Date
- Transaction Type
- Amount Invested

Scheme Performance

Contains:

- Returns
- Risk Metrics
- Expense Ratios

AUM by Fund House

Contains yearly Assets Under Management data.

Monthly SIP Inflows

Contains industry SIP collection data.

Category Inflows

Contains net inflows and outflows by mutual fund category.

Industry Folio Count

Contains monthly investor account statistics.

Benchmark Indices

Contains Nifty benchmark performance.

Portfolio Holdings

Contains sector allocation and portfolio composition.

3. ETL Design and Data Processing

Extract

Raw data was collected from Excel files and imported into Python using Pandas.

Transform

Data cleaning activities included:

- Removing duplicate records
- Handling missing values
- Standardizing column names
- Converting date formats
- Data type validation

Load

Processed datasets were stored as CSV files and loaded into SQLite.

Database Tables

Dimension Table:

- dim_fund

Fact Tables:

- fact_nav
- fact_transactions
- fact_performance
- fact_aum

Database Verification

SQLite validation confirmed successful table creation and data loading.

4. Exploratory Data Analysis (EDA)

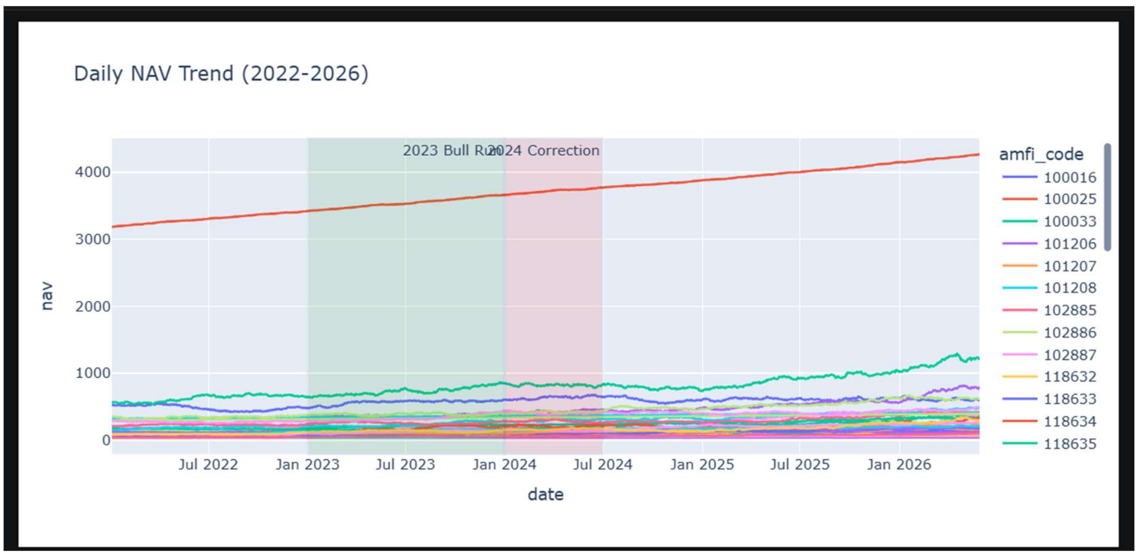
EDA was conducted to understand industry trends and investor behavior.

NAV Trend Analysis

Historical NAV data was analyzed for all schemes between 2022 and 2026.

Key Observation:

- Most equity funds showed strong growth during the 2023 market rally.
- Temporary corrections were observed during 2024.



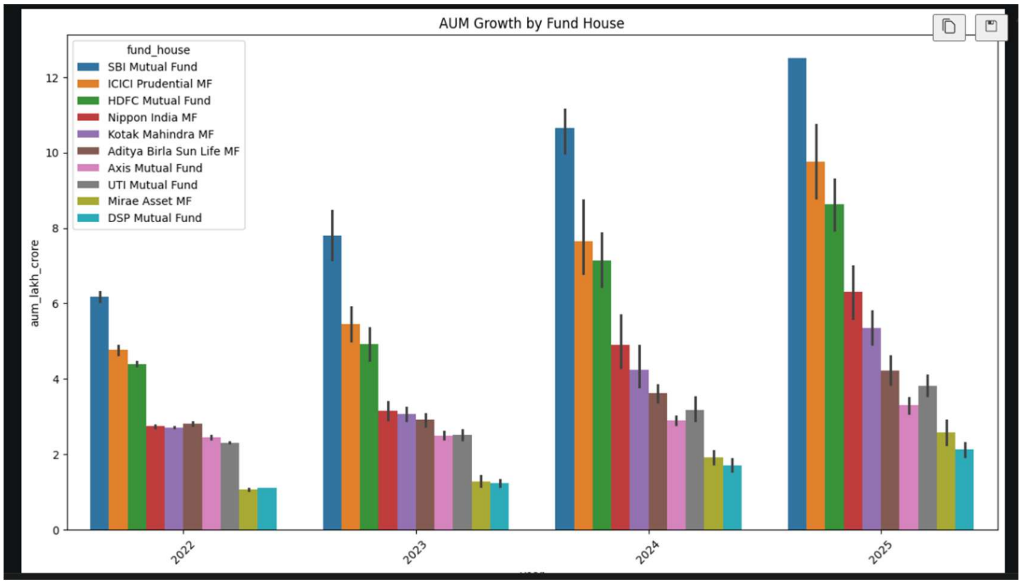
AUM Growth Analysis

AUM growth was analyzed across fund houses.

Key Observation:

- SBI Mutual Fund maintained industry leadership.

- Industry-wide AUM increased significantly during the study period.

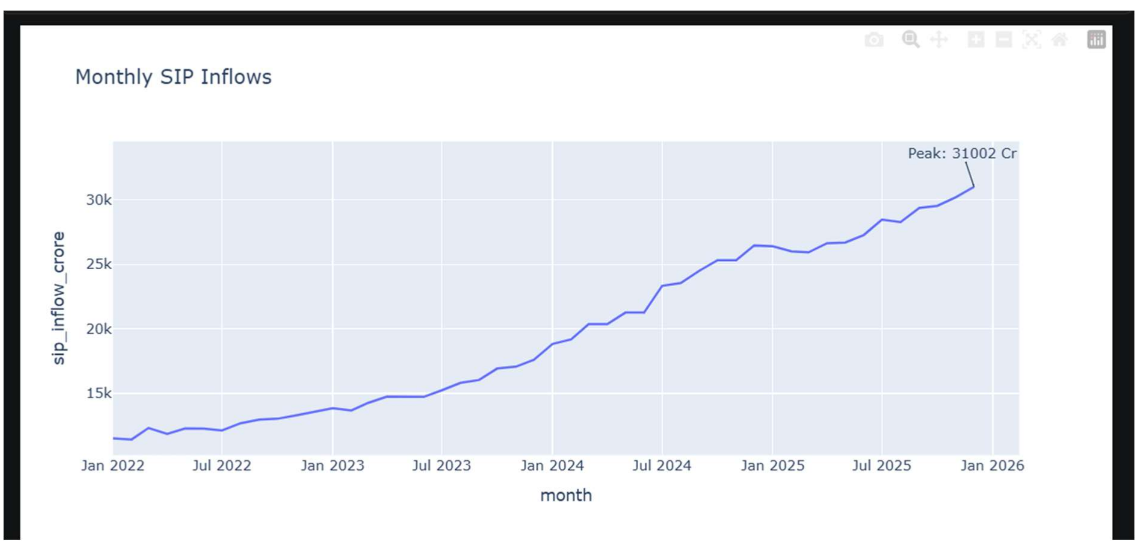


SIP Inflow Analysis

Monthly SIP collections were analyzed.

Key Observation:

- SIP participation increased steadily.
- Industry achieved record monthly inflows by the end of the period.

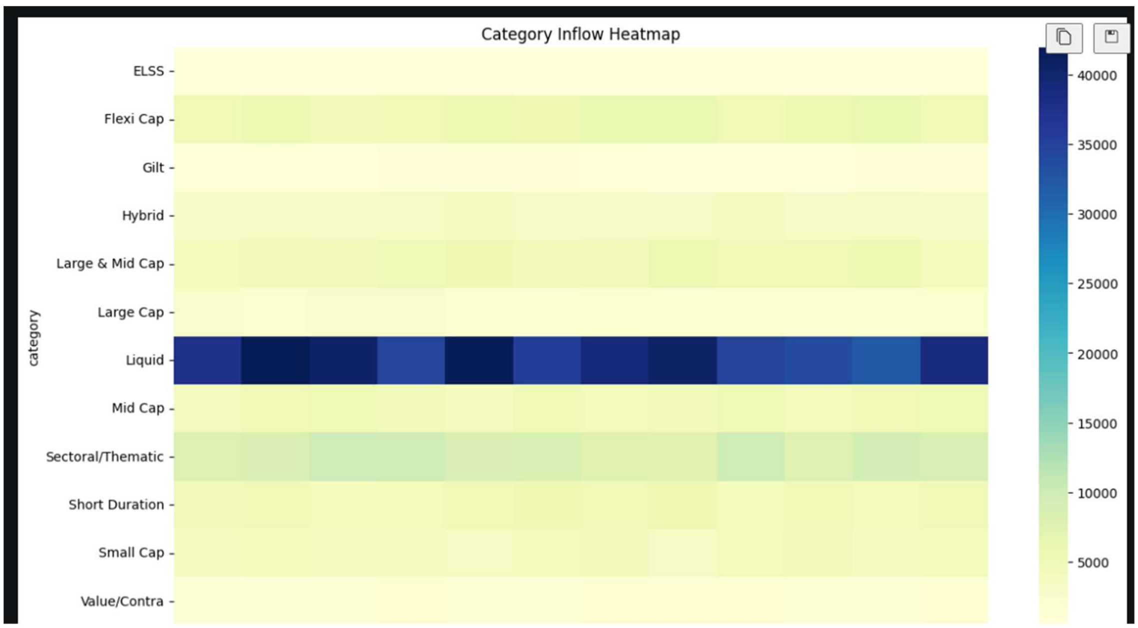


Category Inflow Analysis

Different mutual fund categories exhibited varying inflow patterns.

Key Observation:

- Equity categories attracted the largest inflows.
- Debt categories experienced periodic outflows.



Investor Demographics

Investor participation was analyzed by age group and gender.

Key Observation:

- Middle-aged investors contributed the largest investment volumes.
- SIP participation was highest among salaried individuals.

5. Performance Analytics

CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated for:

- 1 Year
- 3 Year
- 5 Year

The analysis identified funds with consistent long-term growth.

	amfi_code	CAGR
0	100016	2.635246
1	100025	4.455091
2	100033	30.099704
3	101206	23.520489
4	101207	7.933121

Sharpe Ratio

Sharpe Ratio was calculated to measure risk-adjusted returns.

Formula:

Sharpe Ratio = (Portfolio Return − Risk Free Rate) / Standard Deviation

Key Observation:

- Top-performing funds achieved superior risk-adjusted returns.

	amfi_code	Sharpe
34	148567	1.448291
30	120843	1.306744
36	148569	1.234930
19	119551	1.208267
25	120505	1.180101

Sortino Ratio

Sortino Ratio focused specifically on downside risk.

Key Observation:

- Certain funds showed lower downside volatility despite similar returns.

	amfi_code	Sortino
0	100016	-0.351047
1	100025	-0.941821
2	100033	1.829134
3	101206	1.799563
4	101207	0.276644

Alpha and Beta Analysis

Alpha and Beta were calculated against benchmark indices.

Key Observation:

- Positive Alpha funds consistently outperformed benchmarks.
- High Beta funds demonstrated greater market sensitivity.

	amfi_code	alpha	beta
0	100016	0.037476	-0.058268
1	100025	0.042818	0.001158
2	100033	0.271954	0.005104
3	101206	0.213998	0.021086
4	101207	0.108971	-0.065289

Maximum Drawdown

Maximum Drawdown was calculated to identify worst historical losses.

Key Observation:

- Funds with aggressive strategies exhibited deeper drawdowns.

	amfi_code	max_drawdown
0	100016	-0.247344
1	100025	-0.043083
2	100033	-0.162172
3	101206	-0.112916
4	101207	-0.354469

Fund Scorecard

A composite scorecard was developed using:

- Return Rank
- Sharpe Rank
- Alpha Rank

- Expense Ratio Rank
- Drawdown Rank

	amfi_code	scheme_name	fund_score
2	119598	SBI Small Cap Fund - Regular Plan - Growth	82.7500
12	120505	ICICI Pru Midcap Fund - Regular - Growth	81.6250
39	149324	DSP Small Cap Fund - Regular - Growth	77.3125
7	100033	HDFC Mid-Cap Opportunities Fund - Regular - Gr...	75.3750
22	120843	Kotak Flexicap Fund - Regular - Growth	74.6250
26	119094	Axis Midcap Fund - Regular - Growth	69.6250
38	149323	DSP Midcap Fund - Regular - Growth	68.3750
34	148567	Mirae Asset Large Cap Fund - Regular - Growth	68.1250
11	120504	ICICI Pru Bluechip Fund - Direct - Growth	62.3750
36	148569	Mirae Asset Tax Saver Fund - Regular - Growth	62.0625

6. Advanced Analytics

Value at Risk (VaR)

95% Historical VaR was calculated for all schemes.

Purpose:
Estimate maximum expected loss under normal market conditions.

	amfi_code	VaR_95	CVaR_95
0	100016	-0.014364	-0.018060
1	100025	-0.003793	-0.004994
2	100033	-0.019034	-0.023456
3	101206	-0.013282	-0.017439
4	101207	-0.026021	-0.032459

Conditional VaR (CVaR)

CVaR measured average loss beyond the VaR threshold.

Key Observation:

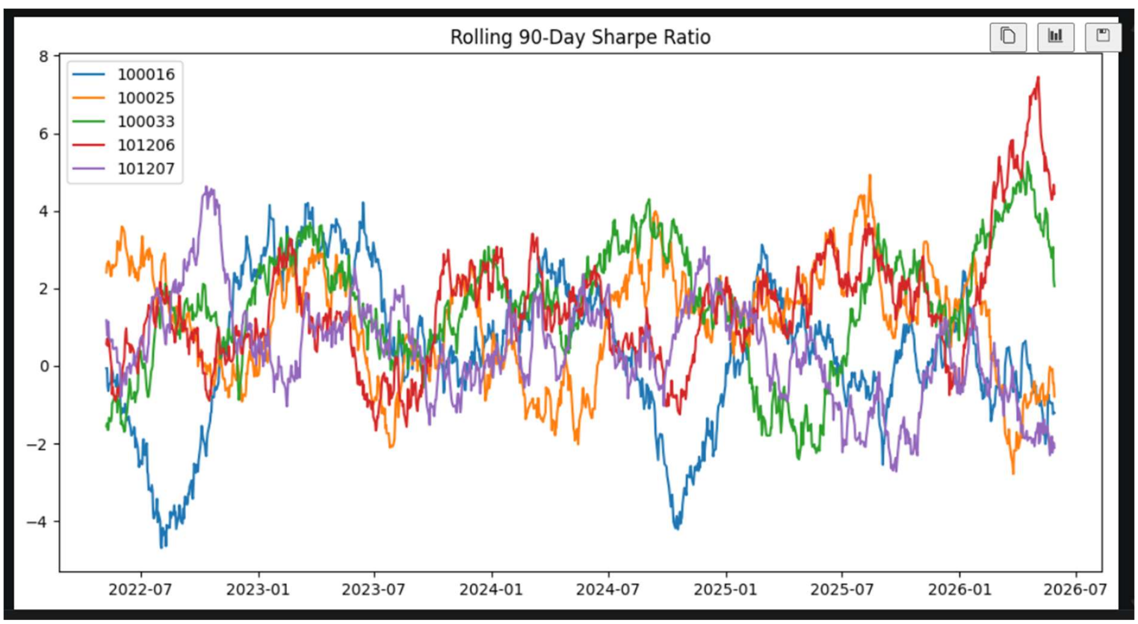
- Certain schemes demonstrated significantly larger tail risks.

Rolling Sharpe Ratio

90-Day Rolling Sharpe Ratios were analyzed.

Key Observation:

- Fund performance varied considerably across market cycles.



Investor Cohort Analysis

Investors were grouped based on first investment year.

Key Observation:

- Recent cohorts invested larger average amounts.

	amount_inr	
	mean	sum
cohort_year		
2024	107422.541832	3491125187
2025	109158.577061	30455243

SIP Continuity Analysis

Investor SIP consistency was evaluated.

Key Observation:

- Most investors maintained regular monthly contributions.
- A subset of investors was identified as at-risk due to missed SIP cycles.

At Risk Investors: 3825

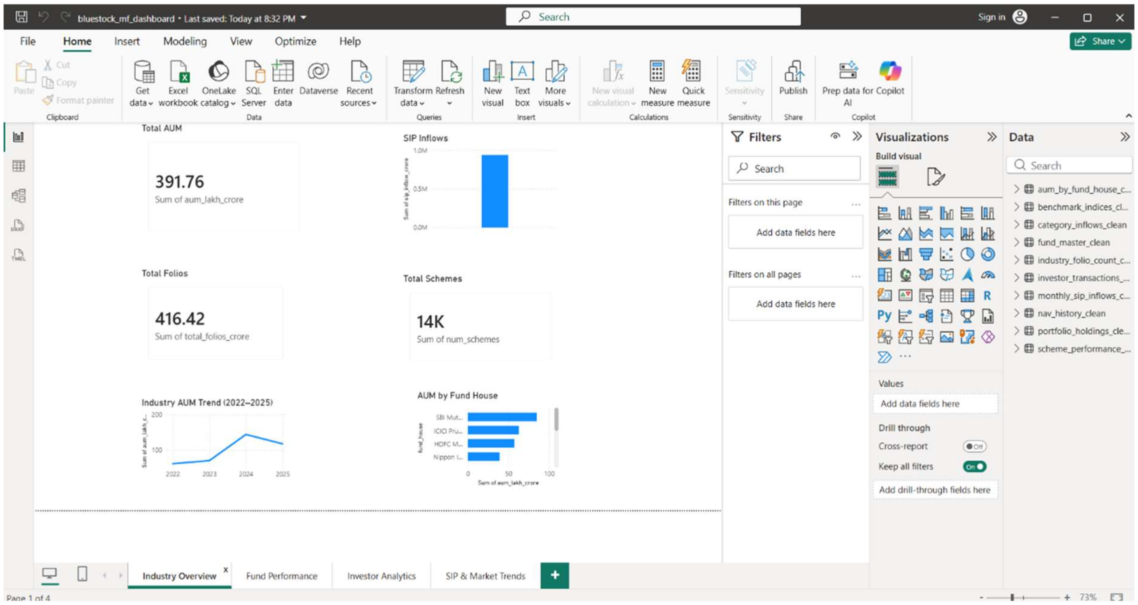
7. Power BI Dashboard

An interactive Power BI dashboard was developed.

Page 1 – Industry Overview

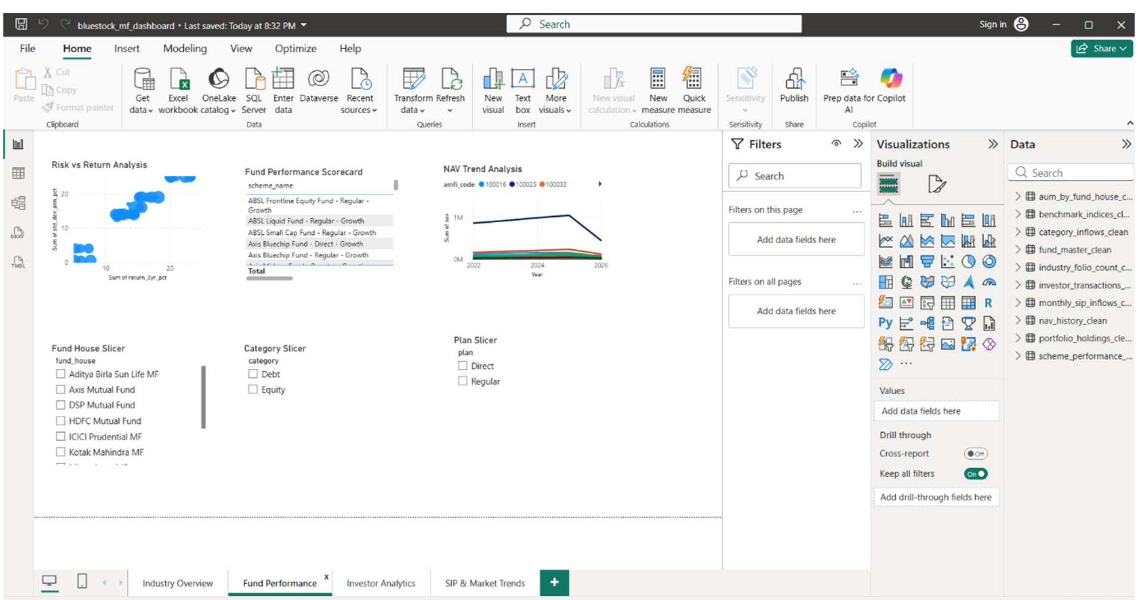
Features:

- Total AUM
- Total Folios
- SIP Inflows
- Industry Growth Trends



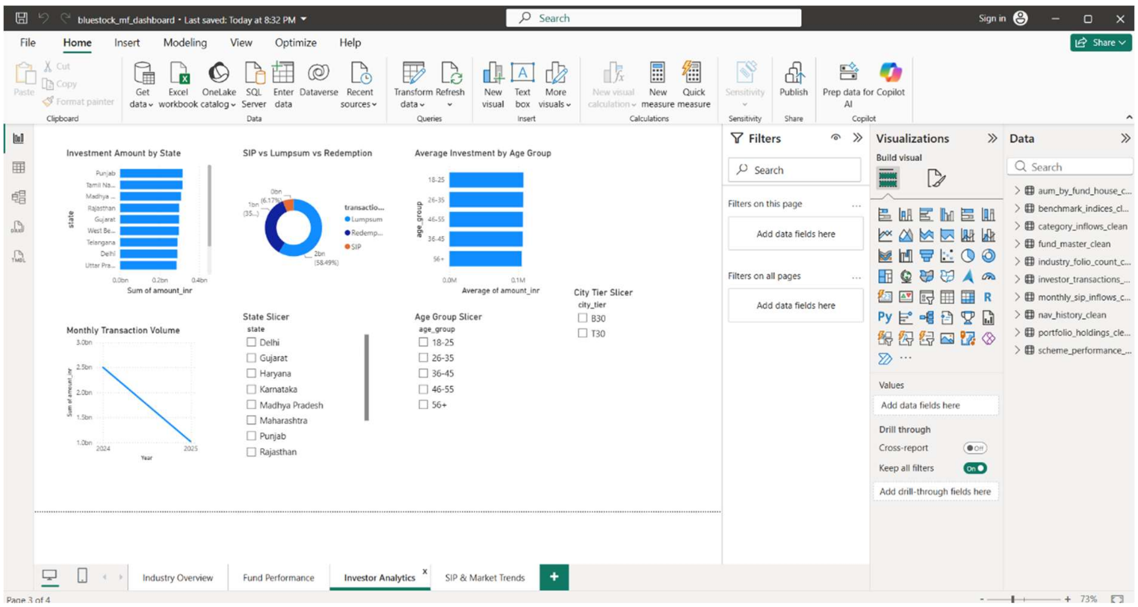
Page 2 – Fund Performance

Features:

- Risk vs Return Scatter Plot
- Fund Scorecard
- NAV Comparison
- 

Page 3 – Investor Analytics

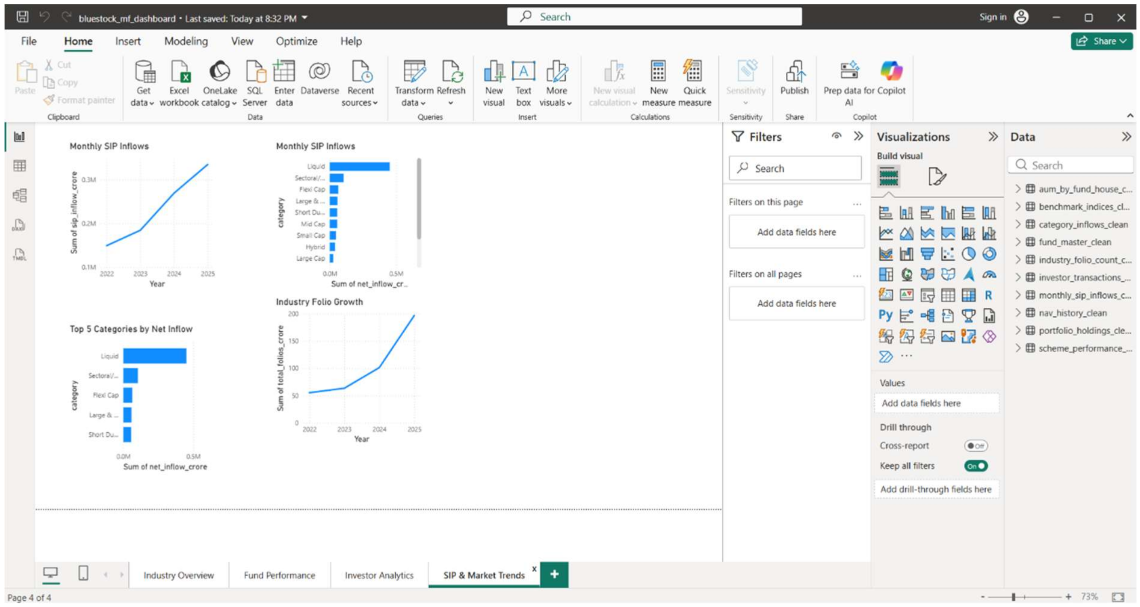
Features:

- State-wise Investments
- Investor Demographics
- Transaction Trends
- 

Page 4 – SIP and Market Trends

Features:

- SIP Growth
- Category Inflows
- Folio Growth



8. Key Findings

1. Industry AUM experienced strong growth throughout the study period.
2. SIP inflows reached record levels.
3. Equity-oriented schemes dominated investor preference.
4. Risk-adjusted performance varied significantly across funds.
5. Positive Alpha funds consistently outperformed benchmarks.

9. Limitations

- Analysis based on historical data only.
- Benchmark coverage limited to available indices.
- Recommendation engine uses simplified scoring.
- Future market conditions may differ significantly.

10. Recommendations

1. Increase investor awareness regarding SIP benefits.
2. Monitor downside risk metrics alongside returns.
3. Diversify portfolios across sectors.
4. Improve investor retention programs.
5. Utilize risk-adjusted metrics for fund selection.

11. Conclusion

The Mutual Fund Analytics Capstone Project successfully delivered a complete analytics solution integrating ETL, database management, statistical analysis, risk evaluation, and dashboard visualization. The project demonstrates how data analytics can support informed investment decisions and provide meaningful insights into mutual fund industry trends.